Hugo Marquerie 26/12/2025

# Caracterización de la continuidad de un operador lineal entre espacios de Banach por elementos del dual

*Observación 1.* Sean $X, Y$ espacios de Banach, entonces, por el `cor-esp-normado-separacion-puntos/??`, podemos tomar $S = Y^*$ (basta con $S = \{L \in Y^* : \|L\| = 1\}$) en el `teo-carac-continuidad-operador-li` y obtener la siguiente caracterización: Sea $T\colon X \longrightarrow Y$ una aplicación lineal, entonces,

$$T\colon X \longrightarrow Y \text{ es continuo } \iff \forall L \in Y^* : L \circ T\colon X \longrightarrow \mathbb{K} \text{ es continuo.}$$

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